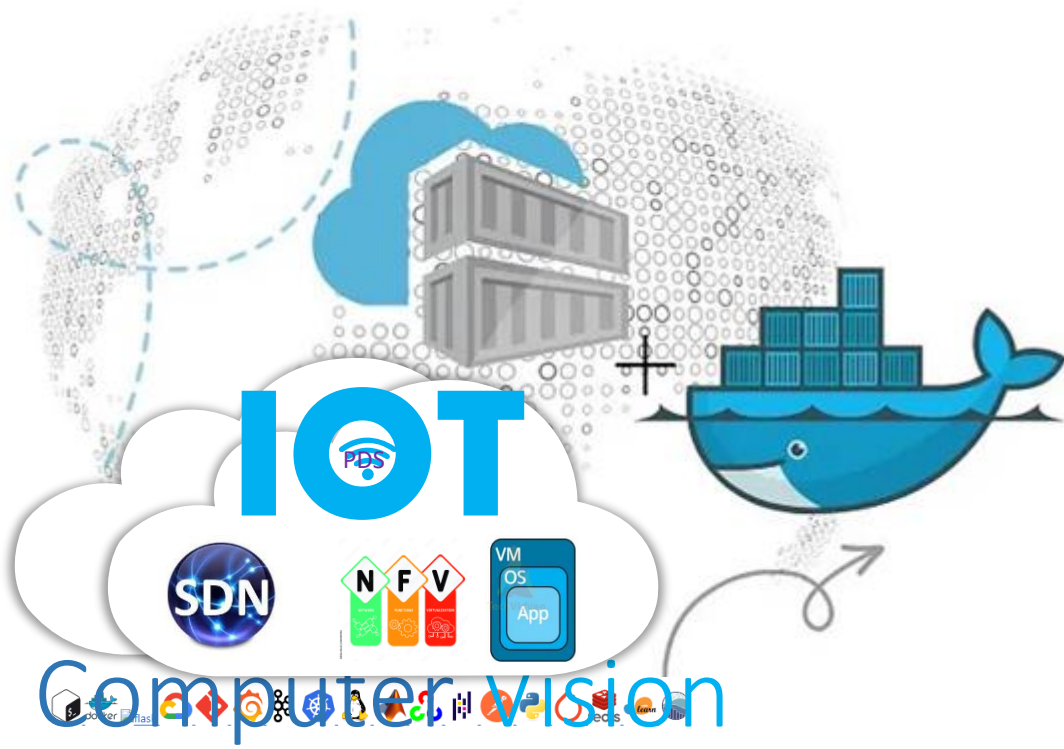
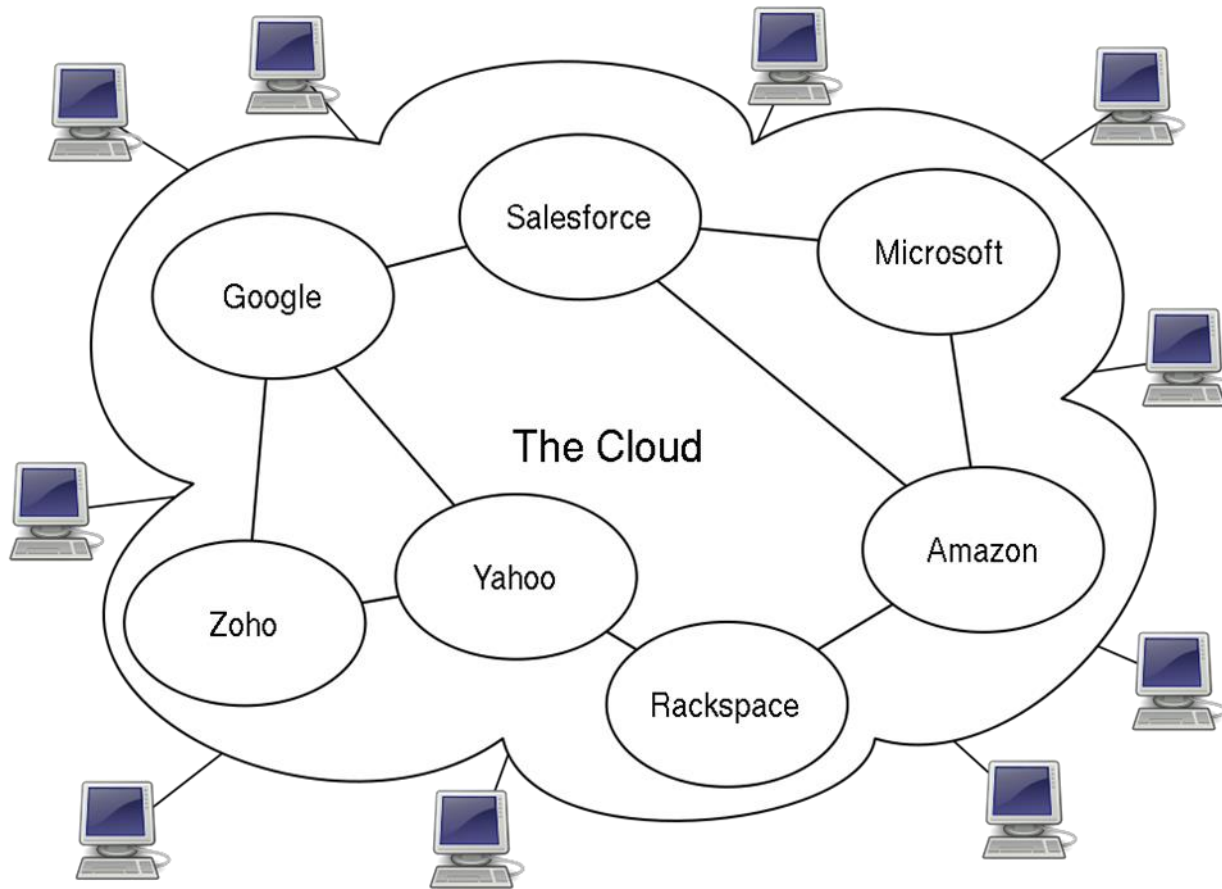




Cloud Computing

<https://github.com/siagianp>
<https://www.youtube.com/@AmelOline/videos>
https://github.com/amelcharolinesgn2/IoT_simulator-mqtt-NodeRed
<https://github.com/amelcharolinesgn2/Cloud-Infrastructures>

Cloud Computing



Definition

The National Institute of Standards and Technology (NIST) provides a concise and specific definition:

- Cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

How it works

- Utilizes the network as a means to connect user end point to resources that are centralized in a data center.
- The data center may be accessed via the internet or a company network, or both.
- Allows access from a variety of end points such as a mobile phone, a PC or a tablet.
- Cloud services may be designed to be vendor agnostic, working equally well with Linux, Mac and PC platforms.

How it works

- Allows access from any internet connected location, allowing Telecommuting, and Outsourcing.
- A user endpoint with minimal software requirements may submit a task for processing. The service provider may pool the processing power of multiple remote computers in "the cloud" to achieve the task, such as data warehousing of hundreds of terabytes, managing and synchronizing multiple documents online, or computationally intensive work.

Comparison

Cloud computing shares characteristics with:

- Autonomic computing
- Client–server model
- Grid computing
- Mainframe computer
- Utility computing
- Peer-to-peer
- Service-oriented computing

Architecture

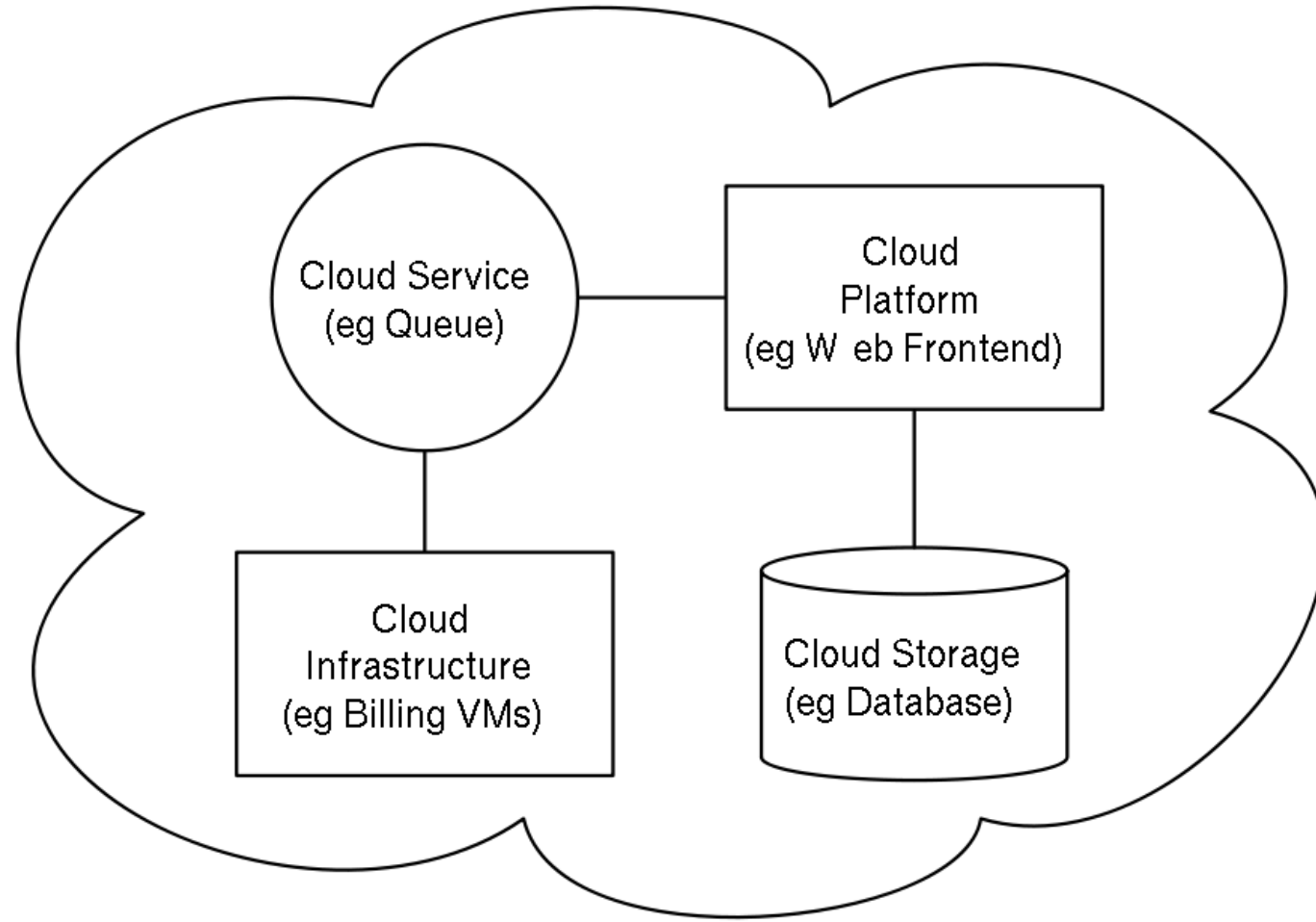
- Involves multiple components communicating with each other over application programming interfaces, usually web services and 3-tier architecture.
- Multiple programs each doing one thing well and working together over universal interfaces.
- Complexity is controlled and the resulting systems are more manageable than their monolithic counterparts.

Architecture

The two most significant components of cloud computing architecture are known as the front end and the back end.

- The front end is the part seen by the client, i.e. the computer user. This includes the client's network and applications used to access the cloud via a user interface such as a web browser.
- The back end of the cloud computing architecture is the 'cloud' itself, comprising various computers, servers and data storage devices.

Architecture



Key characteristics

- Agility improves with users' ability to rapidly and inexpensively re-provision technological infrastructure resources.
- Application Programming Interface (API) accessibility to software that enables machines to interact with cloud software in the same way the user interface facilitates interaction between humans and computers.
- Cost is claimed to be greatly reduced and in a public cloud delivery model capital expenditure is converted to operational expenditure.

Key characteristics

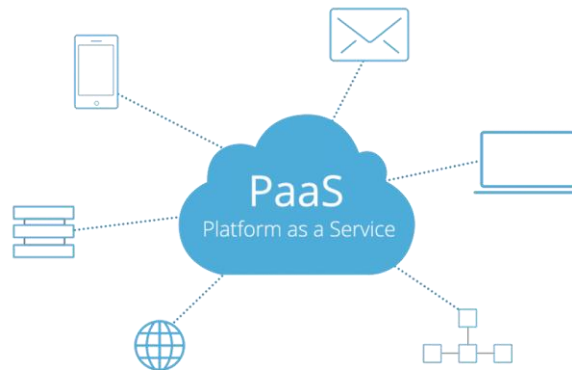
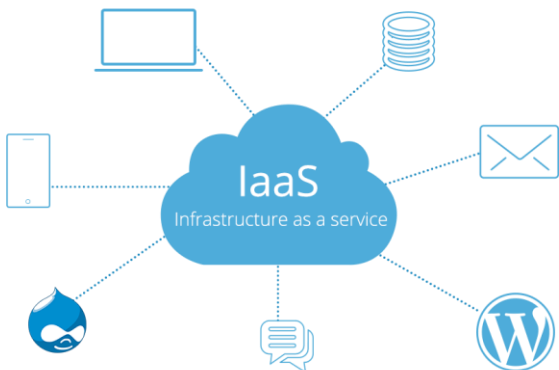
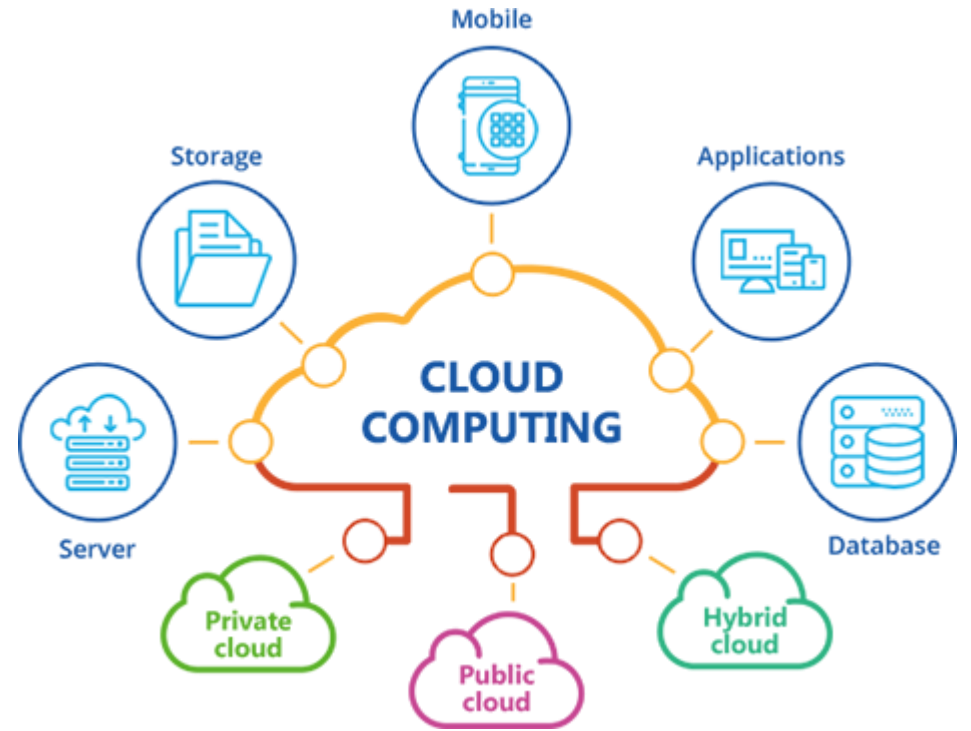
- Device and location independence enable users to access systems using a web browser regardless of their location or what device they are using
- Multi-tenancy enables sharing of resources and costs across a large pool of users thus allowing for:
 - Centralization of infrastructure in locations with lower costs
 - Peak-load capacity increases
 - Utilization and efficiency improvements for systems that are often only 10–20% utilized.

Key characteristics

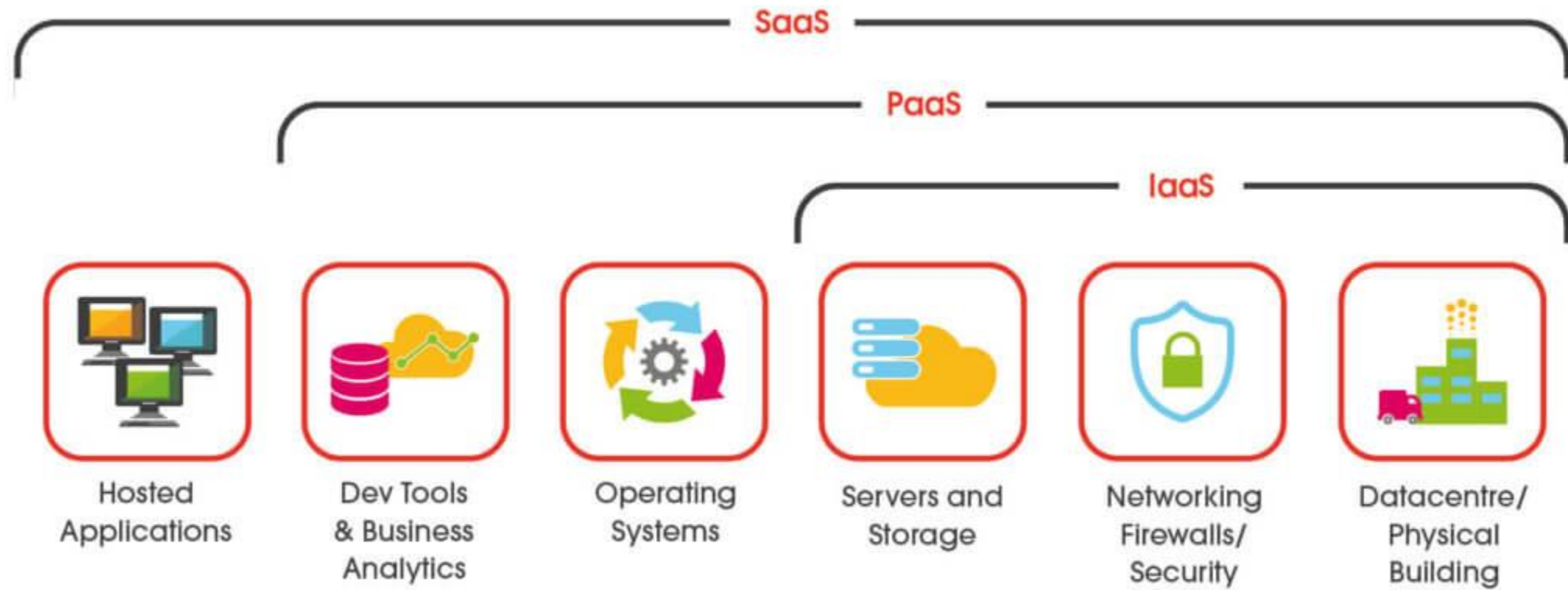
- Reliability is improved if multiple redundant sites are used, which makes cloud computing suitable for business continuity and disaster recovery.
- Scalability via dynamic "on-demand" provisioning of resources on a fine-grained, self-service basis near real-time.
- Performance is monitored, and consistent and loosely coupled architectures are constructed using web services as the system interface.
- Security could improve due to centralization of data.

Service Models

- There are three types of the cloud:
- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)

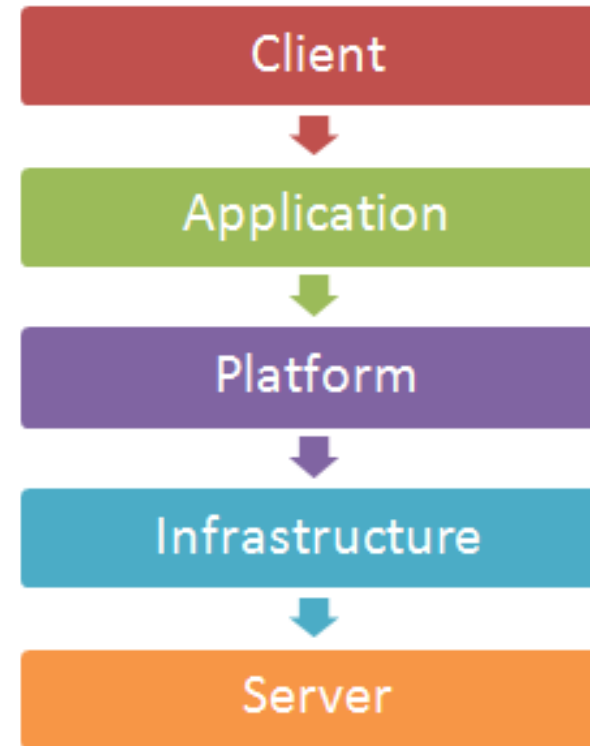


Service Models



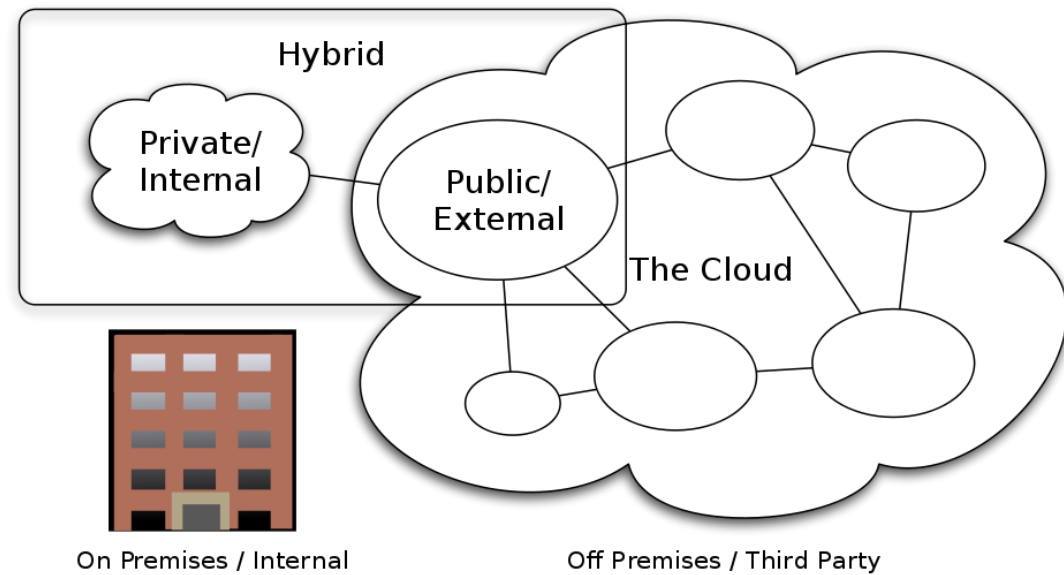
Layers

- Client
- Application
- Platform
- Infrastructure
- Server



Deployment models

- Public cloud
- Community cloud
- Hybrid cloud and hybrid IT delivery
- Combined cloud
- Private cloud



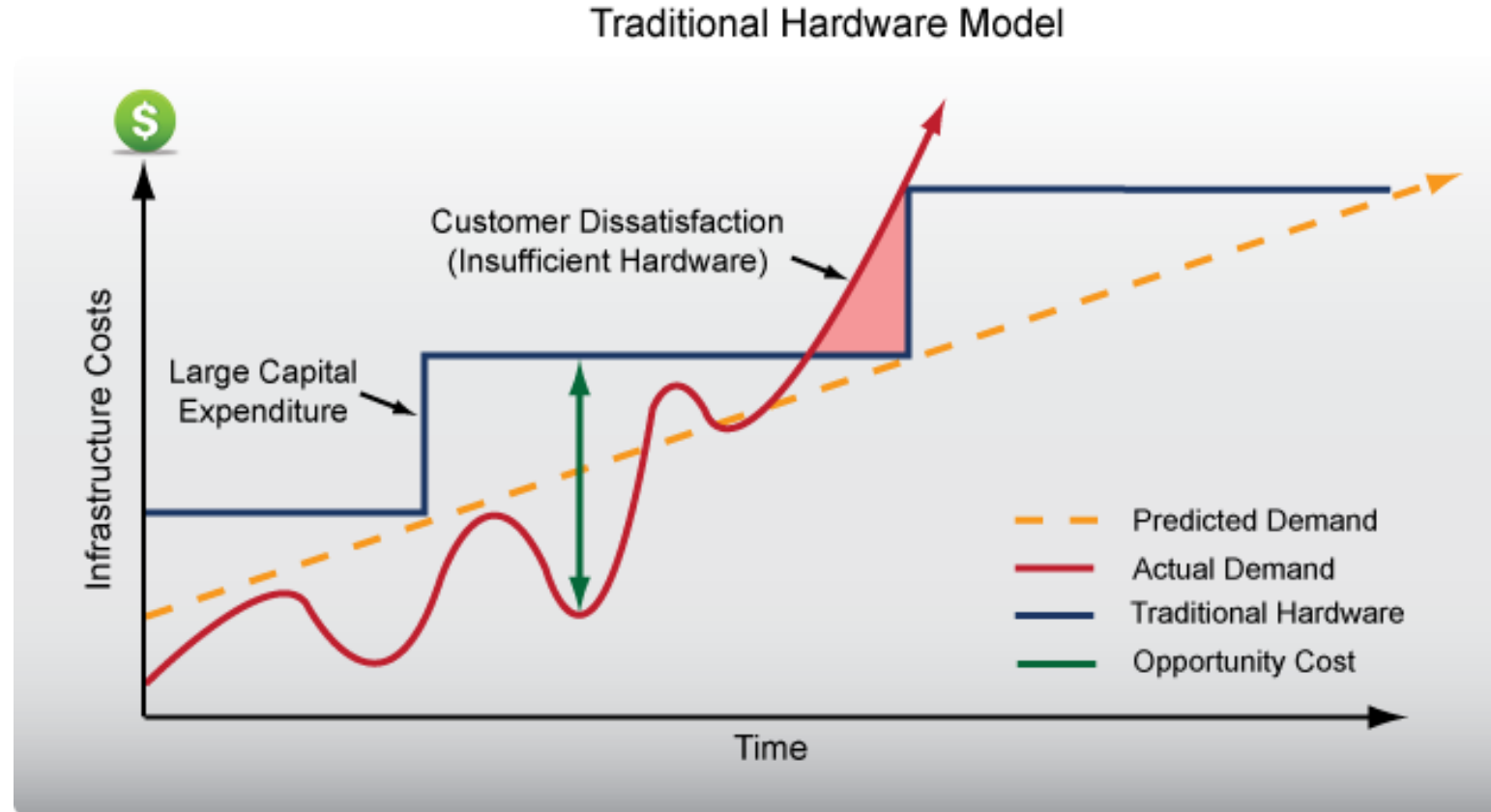
Cloud Computing Types

Why use cloud computing

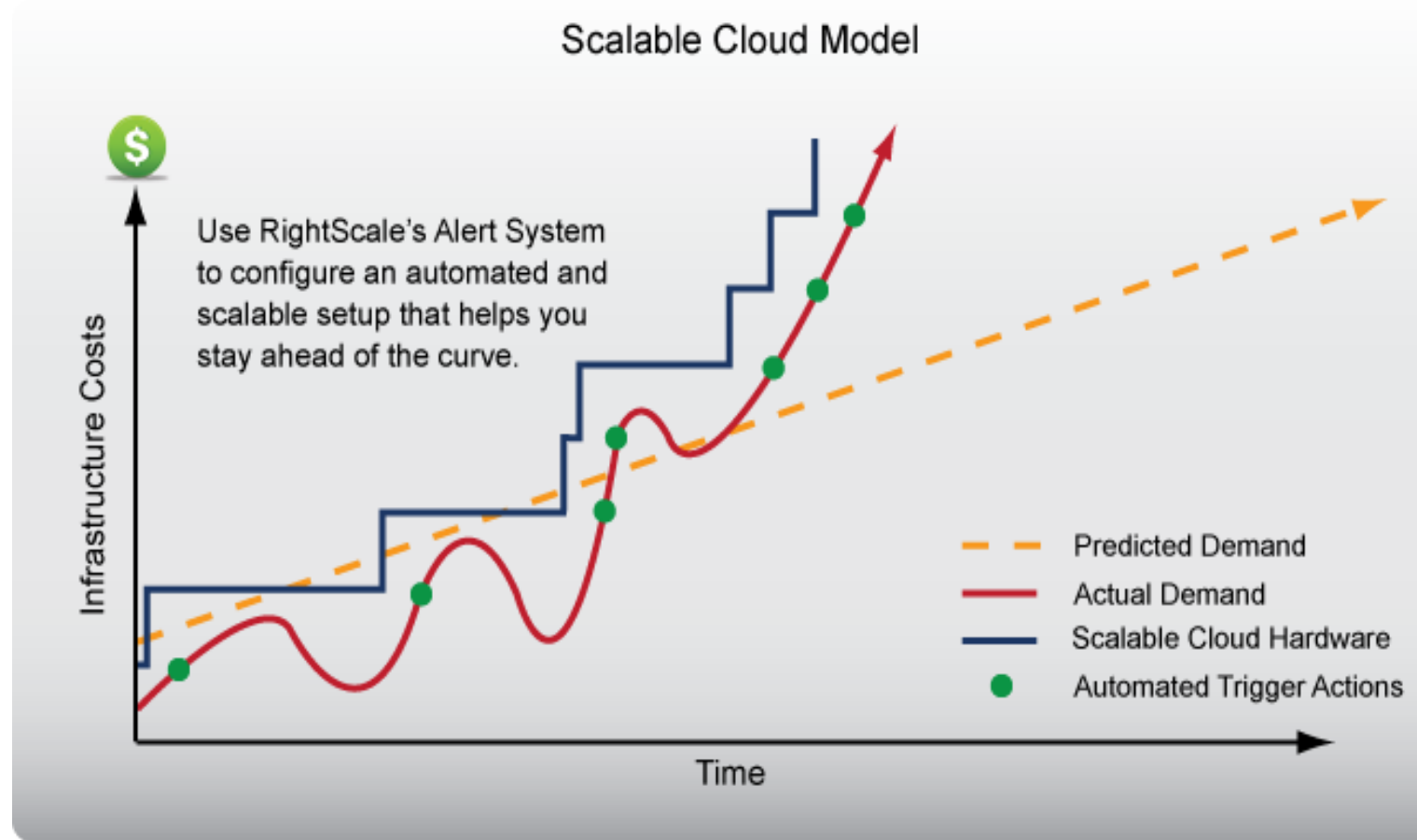
- Website traffic fluctuates.
- Visitors expect steady, fast performance regardless.
- An infrastructure able to handle peak traffic requires capital investment.
- No matter how big you build it, it still may not be enough.
- Having to choose between overspending or under-provisioning is a classic dilemma.

By deploying your web application in the cloud, you can avoid this dilemma. You can eliminate the upfront fixed investment, and still gain access to compute resources to meet peak loads with on demand scalability, while still paying only for the compute time you actually use.

Why use cloud computing



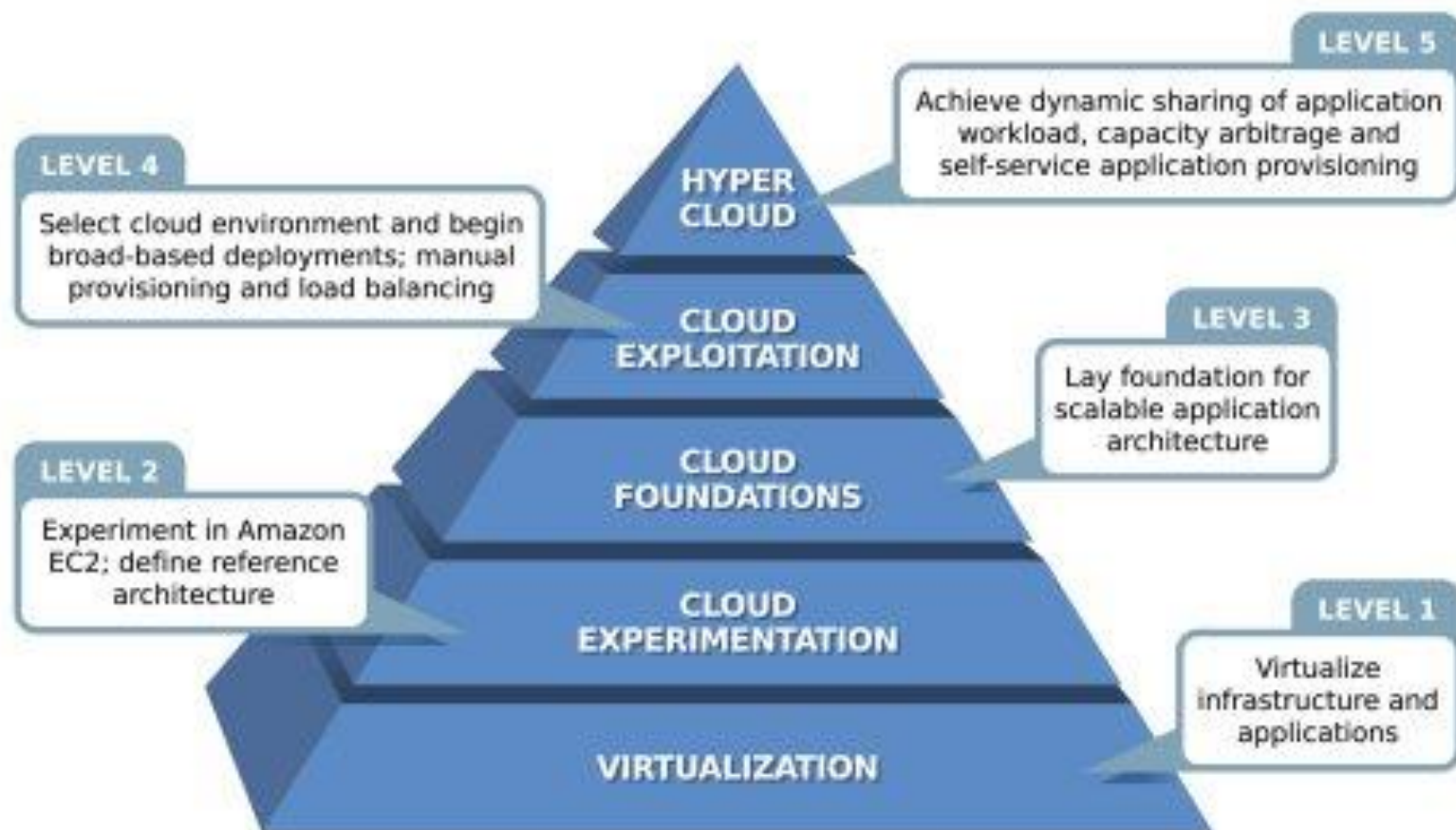
Why use cloud computing



Pros and Cons

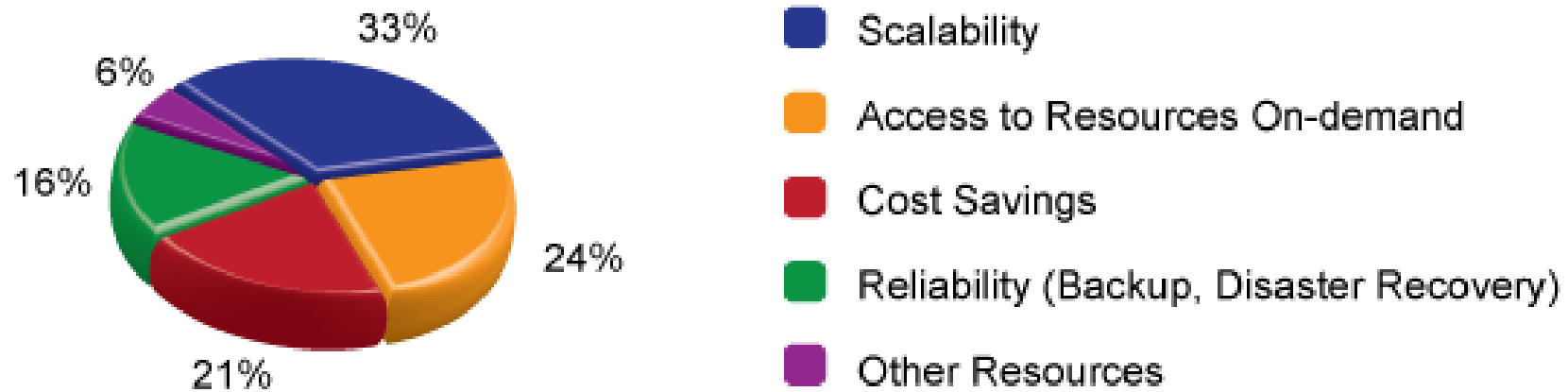


THE CLOUD COMPUTING ADOPTION MODEL



Selection Factors

What drives you to the cloud?



Main Concerns

What are your main concerns?



Questions?



Referensi

- <https://docs.openstack.org/liberty/install-guide-rdo/overview.html>
- <https://compas.cs.stonybrook.edu/cse356-v3>
- <https://compas.cs.stonybrook.edu/~mferdman/#teaching>